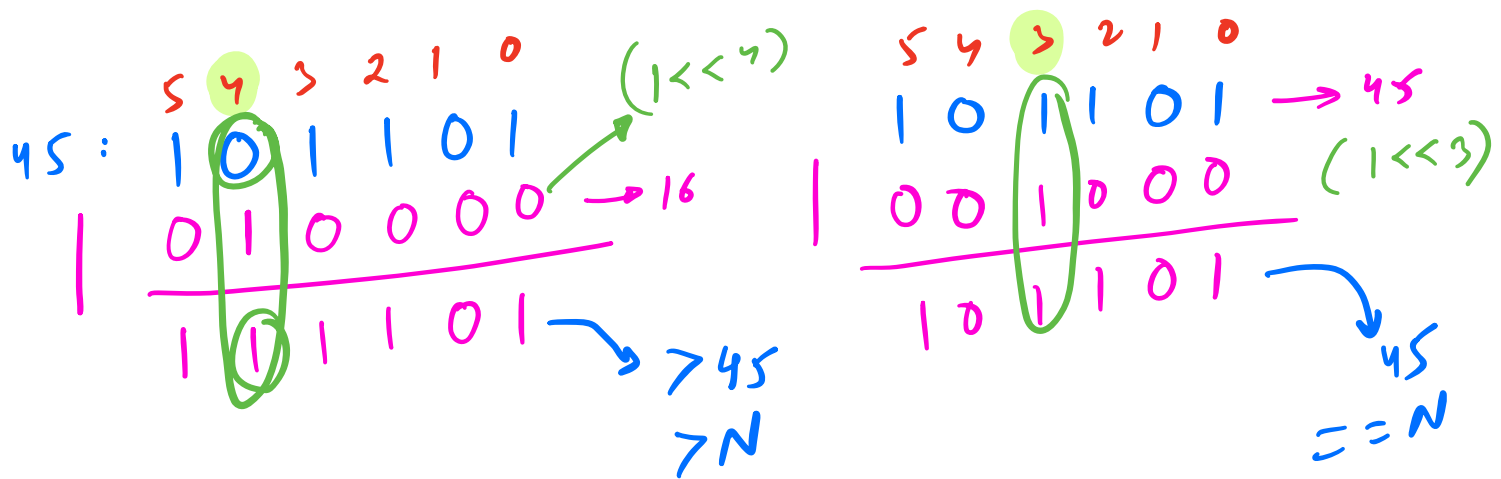
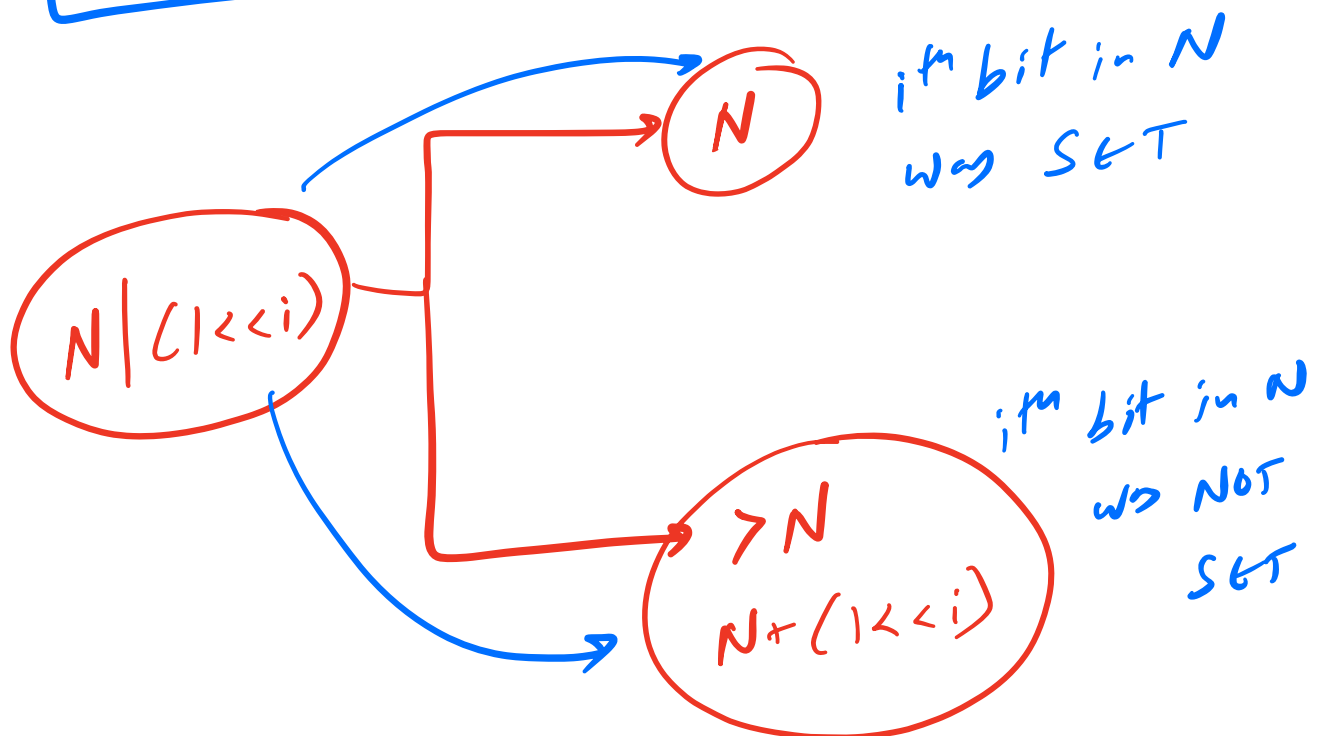


$$N = 45 : 101101$$



Q → Given N , set its i th bit!

$N | (1 \ll i) \rightarrow$ this is N with i th bit as SET



$$\begin{array}{r}
 \begin{array}{cccccc}
 5 & 4 & 3 & 2 & 1 & 0 \\
 45: & 1 & 0 & 1 & 1 & 0 & 1 \\
 \wedge & 0 & 0 & 1 & 0 & 0 & 0
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 1 & 0 & 1 & 0 & 1 & : < N
 \end{array}$$

Bit 3 is circled in pink.

$$\begin{array}{r}
 \begin{array}{cccccc}
 5 & 4 & 3 & 2 & 1 & 0 \\
 45: & 1 & 0 & 1 & 1 & 0 & 1 \\
 \wedge & 0 & 1 & 0 & 0 & 0 & 0
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 1 & 1 & 1 & 1 & 0 & 1
 \end{array}$$

Bit 4 is circled in green.

$(N \wedge (1 \ll i)) \rightarrow$ N with the i^{th} bit toggled

0 $\rightarrow$ 1
 1 $\rightarrow$ 0

$$\begin{array}{r}
 \begin{array}{cccccc}
 5 & 4 & 3 & 2 & 1 & 0 \\
 45: & 1 & 0 & 1 & 1 & 0 & 1 \\
 \wedge & 0 & 0 & 0 & 1 & 0 & 0
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 0 & 0 & 0 & 1 & 0 & 0
 \end{array}$$

Bit 2 is circled in pink.

$$\begin{array}{r}
 \begin{array}{cccccc}
 5 & 4 & 3 & 2 & 1 & 0 \\
 45: & 1 & 0 & 1 & 1 & 0 & 1 \\
 \wedge & 0 & 1 & 0 & 0 & 0 & 0
 \end{array}
 \end{array}$$

$$\begin{array}{r}
 0 & 0 & 0 & 0 & 0 & 0
 \end{array}$$

Bit 4 is circled in pink.

$N \& (1 \ll i) \rightarrow > 0$: i^{th} bit in N is SET

$= 0$: i^{th} bit in N is UNSET

Check if i th bit is SET $\longrightarrow$ AND
 Set the i th bit $\longrightarrow$ OR
 Toggle the i th bit $\longrightarrow$ XOR

Q Given N , UNSet the i th bit!

$N = 25$

$$\begin{array}{r}
 \begin{array}{c} 4 \quad 3 \quad 2 \quad 1 \quad 0 \\ 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ \wedge \quad 0 \quad 1 \quad 0 \quad 0 \quad 0 \\ \hline 1 \quad 0 \quad 0 \quad 0 \quad 1 \end{array}
 \end{array}$$

$$\begin{array}{r}
 \begin{array}{c} 4 \quad 3 \quad 2 \quad 1 \quad 0 \\ 1 \quad 1 \quad 0 \quad 0 \quad 1 \\ \wedge \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \\ \hline 1 \quad 1 \quad 1 \quad 0 \quad 1 \end{array}
 \end{array}$$

X

```

if ((N & (1 << i)) > 0) {
    N = (N ^ (1 << i));
}
return N;
  
```

④

$$\begin{array}{r}
 \begin{array}{cccccc}
 5 & 4 & 3 & 2 & 1 & 0 \\
 1 & 0 & 1 & 1 & 0 & 1 \\
 0 & 0 & 1 & 0 & 0 & 0 \\
 \hline
 1 & 0 & 1 & 1 & 0 & 1 \\
 0 & 0 & 1 & 0 & 0 & 0 \\
 \hline
 1 & 0 & 0 & 1 & 0 & 1
 \end{array}
 \end{array}$$

$(1 \ll i)$
 i^{th} bit is SET

$$N = (N \mid (1 \ll i)) \wedge (1 \ll i)$$

⑤

$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 0 & 1 & 0 \\
 \hline
 1 & 1 & 0 & 1 \\
 \hline
 1 & 0 & 0 & 1
 \end{array}
 \end{array}
 \rightarrow N$$

$$\begin{array}{r}
 \begin{array}{cccc}
 1 & 1 & 1 & 1 \\
 \hline
 1 & 0 & 0 & 1
 \end{array}
 \end{array}
 \rightarrow \sim(1 \ll i)$$

$$N = N \& (\sim(1 \ll i))$$

Q Given N & i .
check if the i th bit is set or NOT!

1) if $((N \gg i) \& 1) == 1$
 $\rightarrow$ i th bit is SET

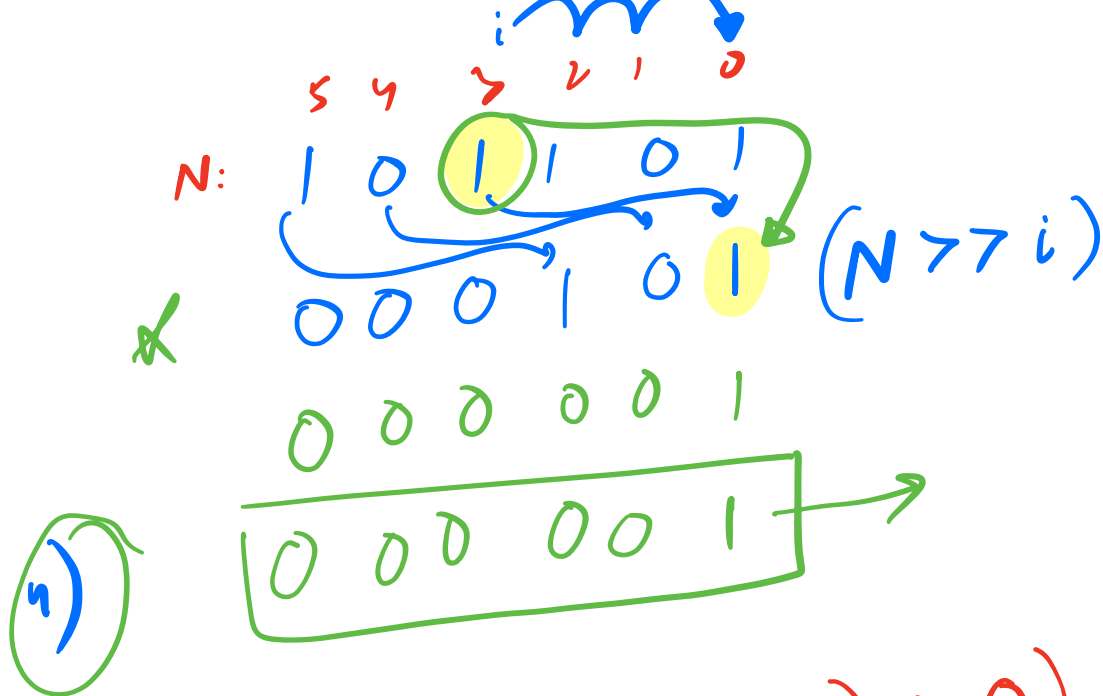
else
 $\rightarrow$ i th bit is UNSET

2) if $((N \gg i) \& 1) > 0$
 $\rightarrow$ i th bit is SET

else
 $\rightarrow$ i th bit is UNSET

3) if $((N \gg i) \& 1) > N$
 $\rightarrow$ i th bit is UNSET

else
 $\rightarrow$ SET



if $((N \gg i) \& 1) > 0$
 ith bit is SET

else $\rightarrow$ NOT!

Q Given a no. N. Count the total no. of set bits!

N = 6 $\rightarrow$ 110 $\rightarrow$ 2

N = 25 $\rightarrow$ 11001 $\rightarrow$ 3

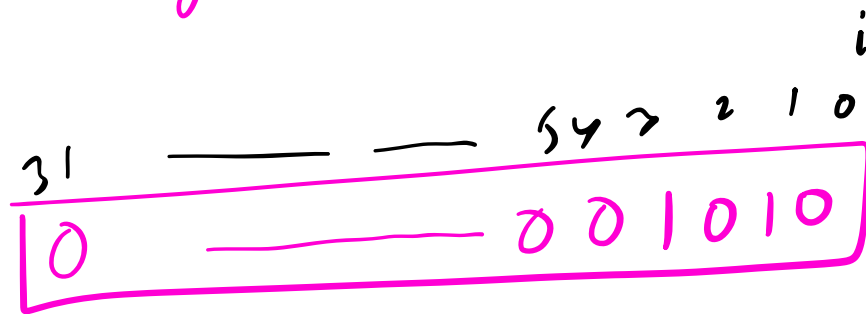
N = 45 $\rightarrow$ 101101 $\rightarrow$ 4

N = 32 $\rightarrow$ 100000 $\rightarrow$ 1

N = 0 $\rightarrow$ 0000 $\rightarrow$ 0

⑥ INT $\rightarrow$ 4 Bytes : 32 bits

$N = 10$



```

cnt = 0;
for (i = 0; i < 32; i++) {
    if ((N & (1 << i)) > 0) {
        cnt++;
    }
}

```

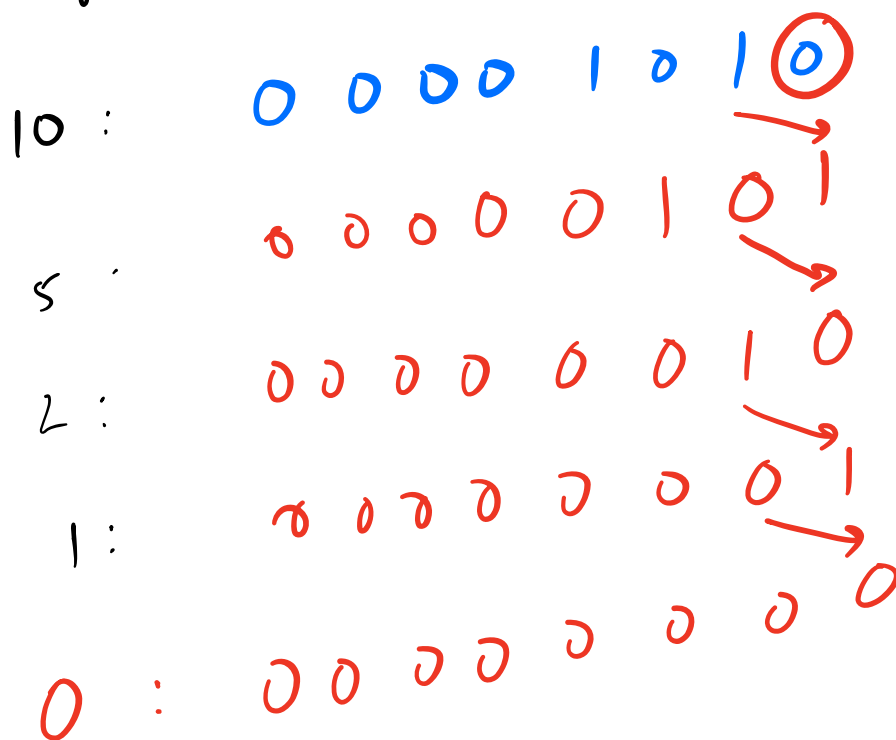
$\rightarrow$ (32)

$\boxed{TC = O(1)}$

32

ret cnt;

cnt = 0



$N = 10$ 0

$N = (N \gg 1)$ 1

$N = (N \gg 1)$ 1

$N = (N \gg 1)$ 2

~~STOP~~

cnt = 0;

```
while (N > 0) {  
    if ((N & 1) > 0) {  
        cnt++;  
    }
```

```
    N = (N >> 1); // N = N/2;
```

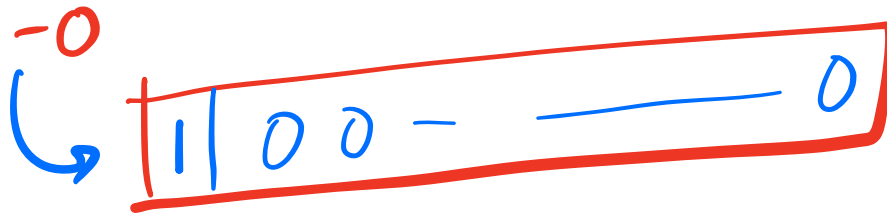
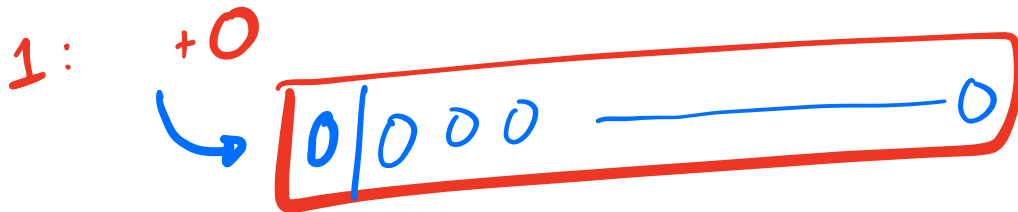
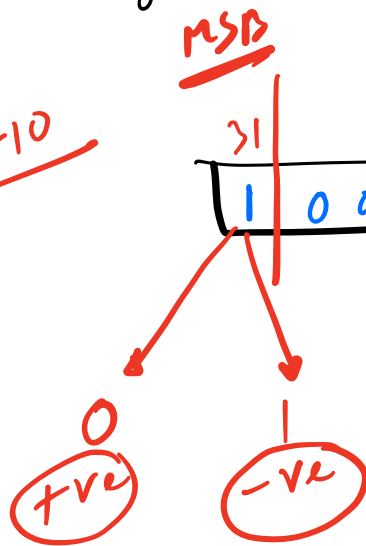
```
}  
return cnt;
```

$TC = O(\log N)$

4

Negative No's →

int x = -10



2 diff. representations for the same no!

2. Additional circuit would be rigid to deal with -ve no's or subtraction!

How are -ve no's stored?

→ 2's complement form

Byte $x = -12$

MSB bit : sign bit
0 : +ve
1 : -ve

12 : 7 6 5 4 3 2 1 0
 0 0 0 0 1 1 0 0
Negation : 1 1 1 1 0 0 1 1 is comp.
(1 → 0)
(0 → 1) + 0 0 0 0 0 0 0 1

(-ve)

-2^7

$$+ 2^6 + 2^5 + 2^4 + 2^2$$
$$-128 + 64 + 32 + 16 + 4$$

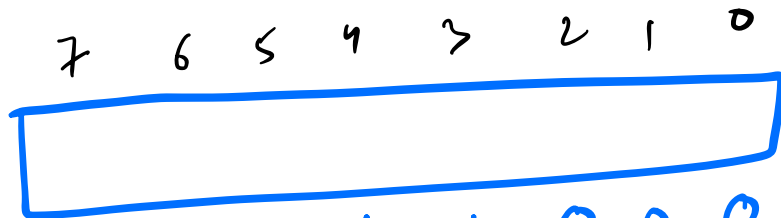
$$-128 + 116$$
$$= -12$$

-12
2's
Comp

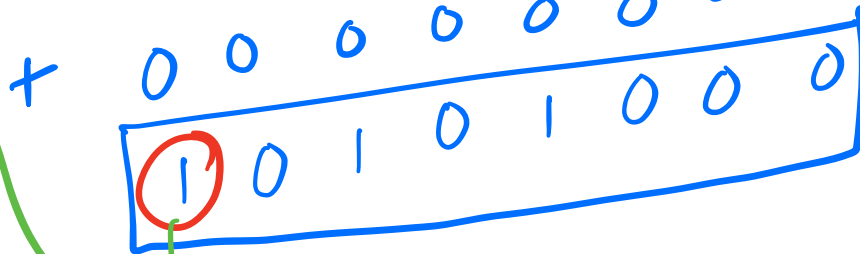
7 6 5 4 3 2 1 0
-2⁷ → 1 0 0 0 0 0 0 0
= -128

8bit signed

byte $x = -88$

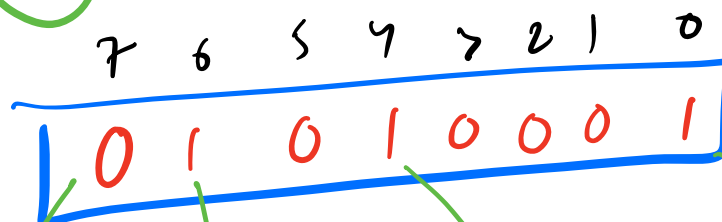


0 1 0 1 1 0 0 0 : 88
 1 0 1 0 0 1 1 1 is -88



-88:
 2's of 88

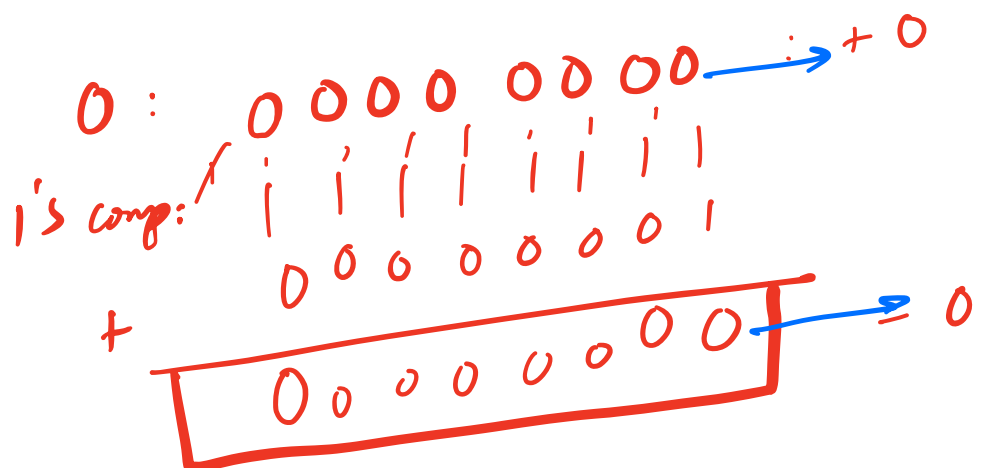
$-2^7 + 2^5 + 2^3$
 $-128 + 40$
 $= -88$



$0 \times -2^7 + 1 \times 2^6 + \dots + 1 \times 2^0$

$2^6 + 2^4 + 2^1 \rightarrow 70$

$N = -0$



8bit signed

$N = 45$

	7	6	5	4	3	2	1	0	
	0	0	1	0	1	1	0	1	
	1	1	0	1	0	0	1	0	1's
	+	0	0	0	0	0	0	1	
	1	1	0	1	0	0	1	1	2's
-45	1	1	0	1	0	0	1	1	
+45	+	0	0	1	0	1	1	0	
=	0	0	0	0	0	0	0	0	

Subtraction of binary No's

$45 - 12 = 45 + (-12) = 33$

45	0	0	1	0	1	1	0	1
+	1	1	1	1	0	1	0	0
-12	0	0	1	0	0	0	0	0
	2 ⁵						2 ⁰	

= 33

→ Subtraction does not need ANY ADDITIONAL circuit. it Uses ADDER CIRCUIT!

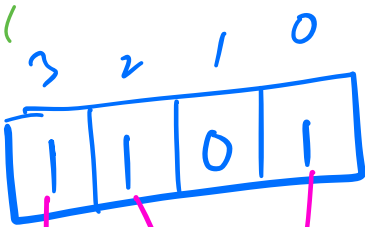
SIGNED

vs

UNSIGNED

$+ve$, $-ve$

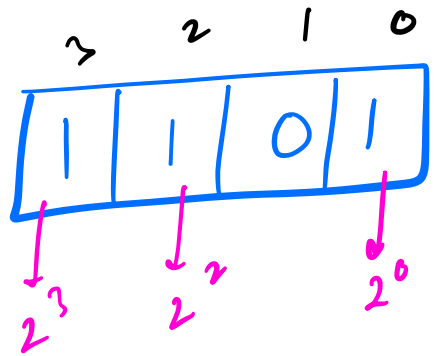
int



$$-2^3 + 2^2 + 2^0 = -3$$

+

Unsigned int

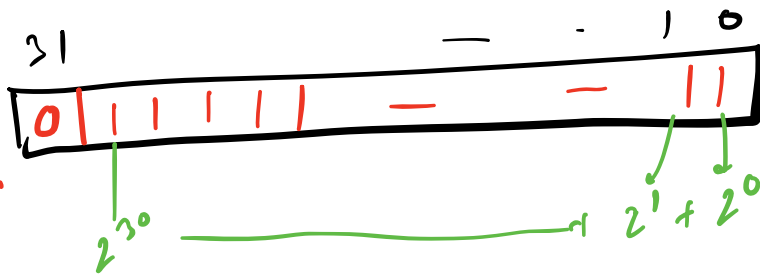


$$8 + 4 + 1 = 13$$



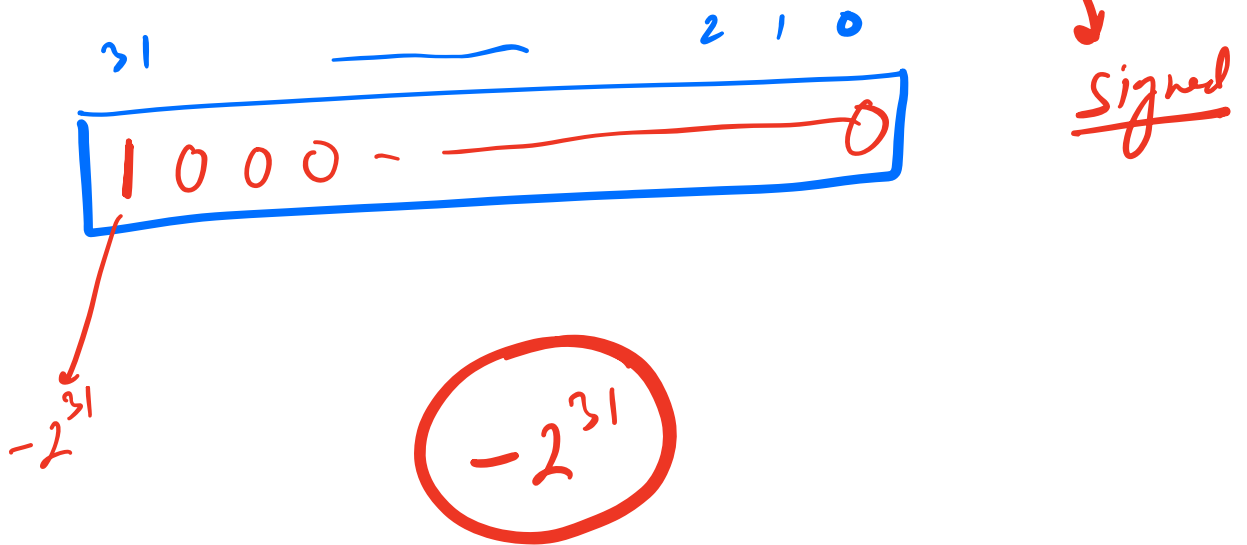
What is the MAX no. on INT can store?

SIGNED



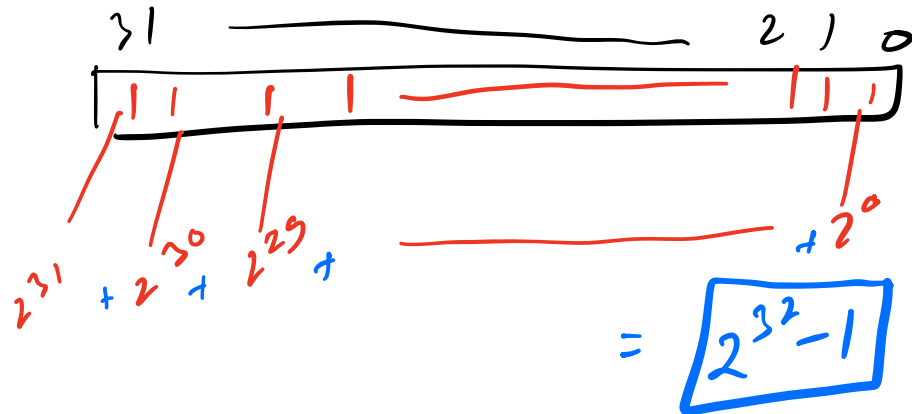
$$= 2^3 - 1$$

Q What is the SMALLEST NO — in INT?

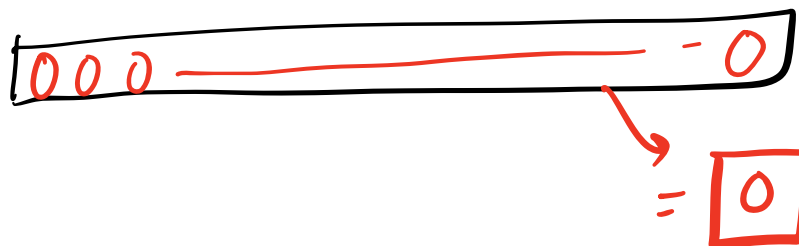


Q → find the range for UNSIGNED INT!

MAX



MIN



Signed

<u>byte</u> 8 bit $2^8 - 1$ 127	<u>int</u> 32 bit $2^{31} - 1$ $\sim 2 \times 10^9$ - $\sim 10^9$ +	<u>long</u> 64 bit $2^{63} - 1$ $\sim 9 \times 10^{18}$ - $\sim 10^{18}$ +
--	--	---

Q Given an array. find the sum of the Array!

~~int~~ sum = 0;

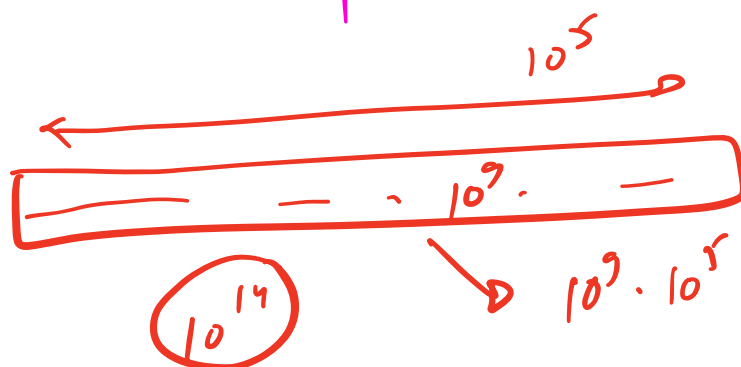
~~long~~ f(i = 0; i < N; i++) {
 sum = sum + A[i];

 }
 return sum;

CONSTRAINTS

$1 \leq N \leq 10^5$

$1 \leq A[i] \leq 10^9$



Q Given 2 ints a & b , ret $a * b$!

$$1 \leq b, a \leq 10^9$$
$$10^9 \times 10^9 = 10^{18}$$

int ans = a * b;

ret ans;

~~overflow~~



long ans = a * b;

ret ans;

~~overflow~~

long ans = long(a * b);

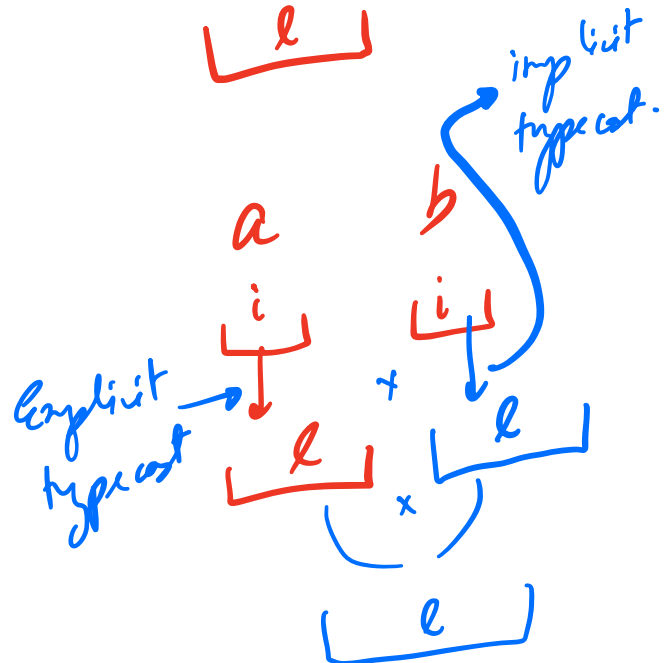
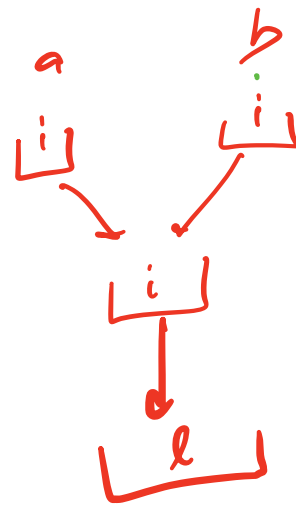
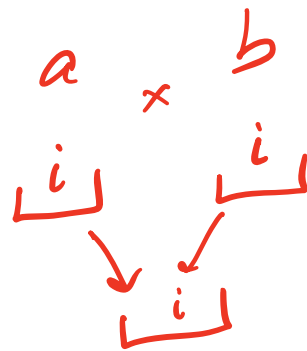
ret ans;

~~or~~

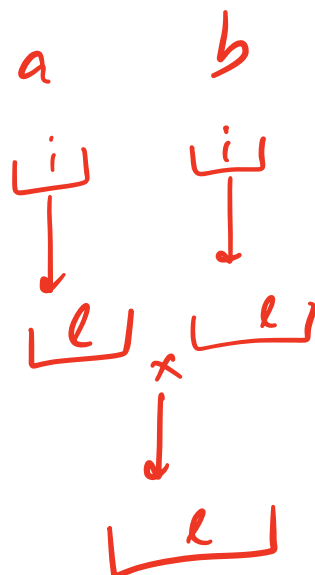
long ans = long(a) * b;

ret ans;

✓

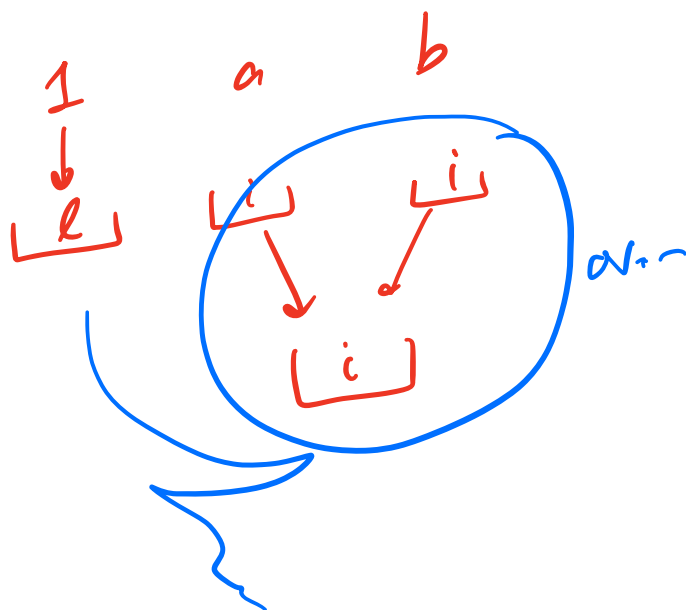


✓ $\log ans = \log(a) + \log(b)$
 set ans;

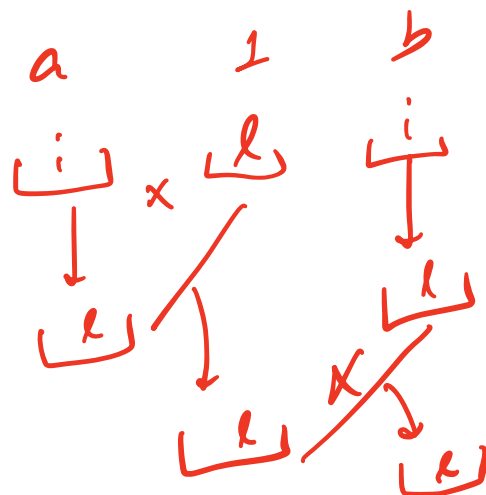


$\log ans = 1L \times a \times b$

✗ set ans;
or



$\log ans = a \times 1L \times b$
 set ans;



1

21 _____ 10
[0 - - - - 0 0 0 1]

[0 - - - 0 0 0 1 0 0 0]

(1) 1 1 1 1 1 0 1 1 1

$1 \ll 3$

$\sim(1 \ll 3)$